

CBSE Class 10 Maths Revision Notes

Chapter 4: Quadratic Equations (Detailed Guide)

1. Introduction & Standard Form

A **quadratic equation** in the variable x is an equation of the form $ax^2 + bx + c = 0$, where a , b , and c are real numbers, and $a \neq 0$.

- a is the coefficient of x^2 (Quadratic coefficient)
- b is the coefficient of x (Linear coefficient)
- c is the constant term

A real number α is called a **root/solution** of the quadratic equation if $a\alpha^2 + b\alpha + c = 0$. A quadratic equation can have at most **two real roots**.

2. Methods of Solving Quadratic Equations

There are two primary methods prescribed in the current Class 10 CBSE syllabus to solve a quadratic equation:

Method A: Factorisation Method (Middle Term Splitting)

In this method, we split the linear term bx into two parts such that their sum equals b and their product equals ac . Then, we take out common factors to rewrite the equation as a product of two linear factors.

Example: Solve $x^2 - 5x + 6 = 0$.

1. Find two numbers that multiply to 6 and add up to -5 $\rightarrow (-2)$ and (-3) .
2. Split: $x^2 - 2x - 3x + 6 = 0$
3. Factorise: $x(x - 2) - 3(x - 2) = 0 \Rightarrow (x - 2)(x - 3) = 0$
4. Roots: $x = 2$ or $x = 3$.

Method B: Quadratic Formula (Sridharacharya's Rule)

For a standard quadratic equation $ax^2 + bx + c = 0$, the roots are given directly by the formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

3. Nature of Roots & The Discriminant

The term inside the square root of the quadratic formula, $D = b^2 - 4ac$, is called the **Discriminant**. It determines the nature of the roots without actually solving the equation:

Condition	Nature of Roots	Graphical Meaning
$D > 0$ ($b^2 - 4ac > 0$)	Two distinct, real roots $x = \frac{-b \pm \sqrt{D}}{2a}$	Parabola cuts the x-axis at two distinct points.

$D = 0$ $(b^2 - 4ac = 0)$	Two equal, real roots $x = -b / 2a$	Parabola touches the x-axis at exactly one point.
$D < 0$ $(b^2 - 4ac < 0)$	No real roots (Imaginary/Complex roots)	Parabola lies completely above or below the x-axis.

4. Key Practice Problems & Word Problem Insights

- **Word Problems on Speed & Distance:** Always form an equation by equating 'Time taken'. Use the framework: *Time = Distance / Speed*. Typically, (Original Time) - (Reduced Time) = Difference in hours.
- **Age Problems:** Represent ages relative to a variable 'x' (present age). Remember, 'n years ago' implies $(x - n)$, and 'n years later' implies $(x + n)$.
- **Consecutive Integers:** Consecutive integers are represented as $x, x+1$. Consecutive odd or even integers are represented as $x, x+2$.